Observed repayment outcomes in LendingClub accepted loans

Group: NullPointer. **Dataset:** user-supplied local LendingClub CSV; original URL unverified.
Code: [Insert repository URL before submission]. README.md accompanies this project.

Name	Roll number	Institute email	Contribution
Satyam Singh	2024514	[To supply]	[To confirm]
Satguru Saran	2024510	[To supply]	[To confirm]
Krishiv Vats	2024308	[To supply]	[To confirm]

1. Dataset

Full CSV audit: 2,260,701 rows and 151 columns; 1,675,133,810 bytes. One row is a source record; inference uses loans with terminal outcomes. Issue-date range: 2007-06-01 to 2018-12-01. The local SHA-256 is saved in outputs/dataset_audit.json. Original acquisition date and licence are unknown.

The supplied profile reports 314,168 rows and 162 columns, with April-December 2015 issues. Its counts do not describe this full CSV. The 115 numeric plus 36 text fields total 151; parsed date fields may explain the profile column difference, but its exact run cannot be reconstructed.

Before encoding: 1,345,310 terminal loans and 12 selected attributes. Complete-case one-hot representation: 1,344,080 x 40 = 53,763,200 instance-attribute cells; meets the 3M requirement. No IDs, outcomes or repayment-process features are counted.

Key attributes. Interest rate and DTI are percentage-point measures; loan amount and annual income are dollar amounts; FICO is an ordered score band; grade is ordinal A–G; term is categorical 36/60 months. Housing tenure, verification, purpose and application type are nominal. Exact vintage definitions remain a provenance limitation.

2. Data preparation

All source columns were audited in 50,000-row chunks. String-normalised hashing locates duplicate candidates, with exact full-row comparisons when candidates exist. IDs are separately checked. Categories are stripped of whitespace; numeric fields and issue months are parsed with logged failures. No outcome or numeric imputation is used. Each test uses its own complete cases.

Only Fully Paid and Charged Off labels define the primary terminal sample. Current, late, grace-period, Default and ambiguous labels are excluded. Do not equate unresolved loans with successful repayment. Missing special-program and joint fields may be structural; global deletion would be inappropriate.

IQR fences flag observations, not errors. All observed extremes remain in primary tests. A pooled 99th-percentile cap is a sensitivity analysis only. Income is plotted on original and log(1+income) scales for nonnegative values; no Gaussianity is claimed. One-hot encoding uses six selected categorical fields and six numeric fields, avoiding high-cardinality IDs/text and future repayment fields. The source contains 32 total-funded footer records and 1 policy-section separator, all with missing loan amount/status; these are not loans and are excluded from every test. There are 2,260,668 loan records. Two observed DTI values are negative and 135 equal 999; their exact semantics are unverified. Primary inference retains them transparently. Excluding the two negative DTI values gives difference 2.35953, CI [2.31024, 2.40882], leaving the interpretation unchanged. There are 1,667 zero incomes and 7,343 utilisation values above 100; these are flagged, not automatically deleted. No negative income was observed.

3. What changed after preparation

2,260,701 source rows = 1,345,310 terminal rows + 915,391 other/missing-status rows. Encoded complete cases retain 1,344,080, excluding 1,230 terminal rows with missing selected fields. Exact duplicates: 0; repeated non-missing IDs: 0. No duplicate-feature records are deleted merely for matching values.

Per-column missingness, outlier counts, parsing changes and before/after dimensions are in the reproducible output tables. The matrix size is an actual constructed representation; test denominators need not match it.

4. Basic questions and problem statement

EDA asks which statuses dominate, where missingness concentrates, and how borrower/loan descriptors vary. Our primary question quantifies the rate difference between terminal outcomes against a declared meaningful threshold. Supporting questions examine DTI, grade and term. This goes beyond descriptive status counts by separating magnitude, uncertainty and practical importance; it is not claimed as a novel discovery in lending research.

5. Statistical inferences

All hypotheses were written before group tests. $\alpha = 0.05$; numeric and proportion alternatives are two-sided. Chi-square uses its upper tail to detect any categorical association. Thresholds are analytical judgments, not industry standards. Loans are treated as independent for inference, an unverified assumption; large samples do not cure selection or dependence. Four tests form one Holm-adjusted family; intervals remain marginal 95% intervals.

5.1 Interest rate

a) Hypotheses and choice. $H_0 : \mu_{CO} - \mu_{FP} = 0; \quad H_1 : \mu_{CO} - \mu_{FP} \neq 0$

Welch t permits unequal unknown variances. A 1 percentage-point difference is the declared threshold (about 100 dollars annual simple interest per 10,000 dollars principal, not amortised cost).

b) Alternatives. Known-sigma z is unsuitable; population SD is unknown. Pooled t adds an unnecessary equal-variance assumption. Paired t lacks pairs. Rank tests target a different distributional question.

c) Result. n(CO)=268,559, n(FP)=1,076,751; excluded terminal rows=0. Estimate 3.0874; 95% CI [3.067, 3.1079]; statistic 295.97; df 390,130; p < 2.23e-308 (floating-point underflow). Reject H0. Holm p < 2.23e-308 (floating-point underflow). CI beyond positive threshold.

Group means 15.711, 12.623; SDs 4.9104, 4.5272; SE 0.010432.

Charged Off loans have higher mean issued rates in this terminal sample, beyond the chosen meaningful-difference threshold.

5.2 Debt-to-income

a) Hypotheses and choice. $H_0 : \mu_{CO} - \mu_{FP} = 0; \quad H_1 : \mu_{CO} - \mu_{FP} \neq 0$

Welch t on the original scale. Threshold: 2 percentage points (100 dollars monthly debt per 5,000 dollars monthly income illustrates scale). Extreme values and missingness warrant sensitivity analysis.

b) Alternatives. Pooled, known-sigma and paired tests have the same limitations as for rate. Log/rank analyses change the estimand; primary inference retains original values.

c) Result. n(CO)=268,488, n(FP)=1,076,448; excluded terminal rows=374. Estimate 2.3596; 95% CI [2.3103, 2.4089]; statistic 93.8212; df 390,672; p < 2.23e-308 (floating-point underflow). Reject H0. Holm p < 2.23e-308 (floating-point underflow). CI beyond positive threshold.

Group means 20.171, 17.812; SDs 11.832, 10.936; SE 0.02515.

Charged Off loans have higher mean DTI in this terminal sample, beyond the chosen meaningful-difference threshold.

5.3 Grade

a) Hypotheses and choice. H_0 : independence; H_1 : association

Pearson chi-square on the grade-by-outcome counts; Cramer's V measures magnitude. V=0.10 is a weak exploratory convention, not a validated business threshold. A 1,000-draw IID multinomial bootstrap gives its interval (extension).

b) Alternatives. A correlation test would impose an arbitrary numeric grade spacing. Sparse-table exact methods are alternatives when expected counts fail; all expected cells are checked. No post-hoc grade tests are counted.

c) Result. N=1,345,310; excluded terminal rows=0. Estimate 0.26191; 95% CI [0.26029, 0.26345]; statistic 92284.3; df 6; p < 2.23e-308 (floating-point underflow). Reject H0. Holm p < 2.23e-308 (floating-point underflow). CI beyond positive threshold.

Minimum expected cell count: 1823.

Grade and terminal outcome are associated; V exceeds the exploratory magnitude threshold. This is not independent causal evidence beyond the rate comparison.

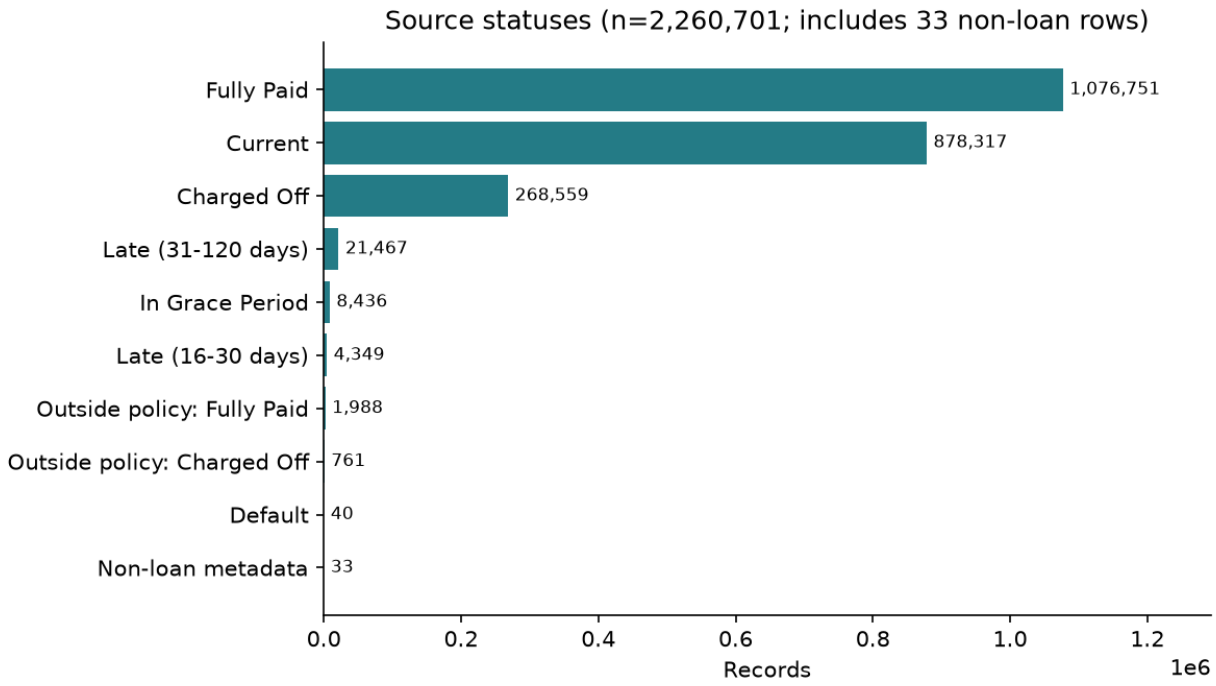


Figure 1: All source statuses. Unresolved loans make terminal-only denominator selection consequential.

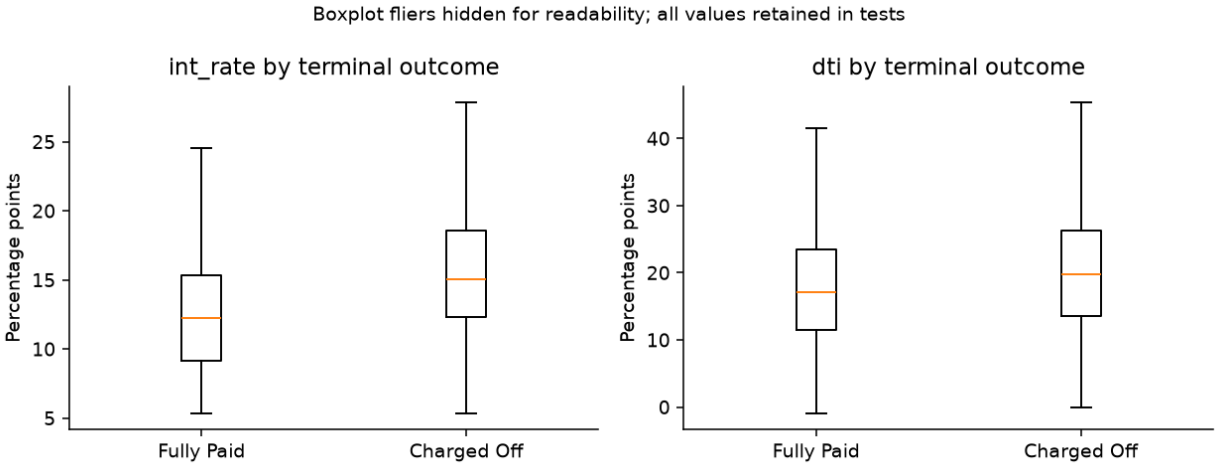


Figure 2: Rate and DTI distributions by terminal outcome (1,345,310 terminal loans before variable-specific missingness). Fliers are hidden only in this plot; full observations enter tests.

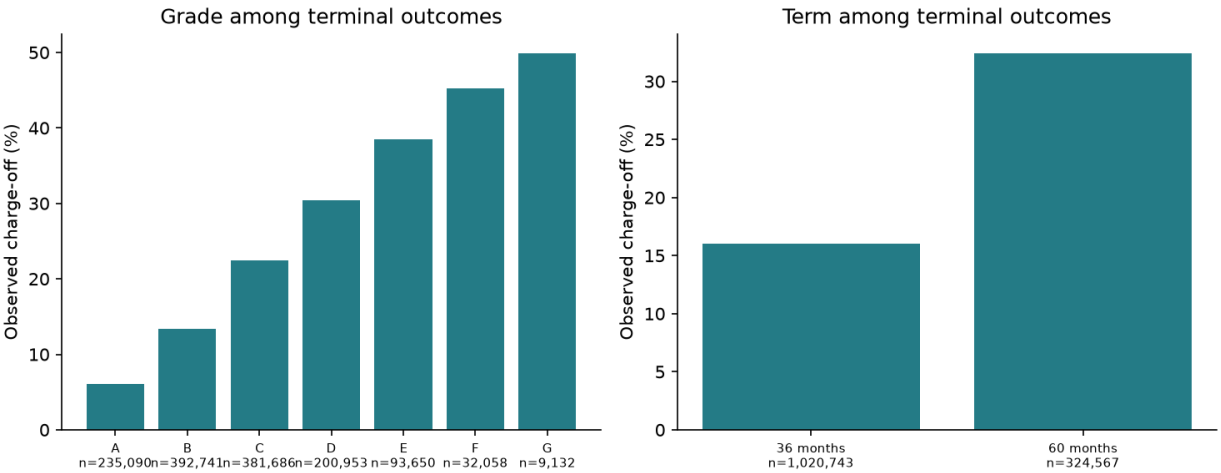


Figure 3: Observed terminal charge-off proportions by grade and term. Composition and outcome maturity can confound these patterns.

5.4 Term

a) Hypotheses and choice. $H_0 : p_{60} - p_{36} = 0; \quad H_1 : p_{60} - p_{36} \neq 0$

Pooled-null two-proportion z; unpooled large-sample difference interval. Event = Charged Off, denominator = terminal loans of each term. Threshold .02 means 2 additional events per 100 terminal loans.

b) Alternatives. The uncorrected 2x2 chi-square is equivalent (z squared), checked but not counted as another finding. Fisher exact is unnecessary when expected cells are adequate. A t test is not the natural binary-count method.

c) Result. n(60)=324,567, n(36)=1,020,743; excluded terminal rows=0. Estimate 0.16452; 95% CI [0.16276, 0.16628]; statistic 204.25; df not applicable; p < 2.23e-308 (floating-point underflow). Reject H0. Holm p < 2.23e-308 (floating-point underflow). CI beyond positive threshold.

Minimum expected cell count: 64792.

Observed event proportions: 0.32445 (60 months), 0.15993 (36 months). Difference/interval above are proportions, not percentage points.

The 60-month terminal group has a higher observed charge-off proportion by about 16.45 percentage points. Selection and maturity prevent a causal interpretation.

Calculation audit. Welch: $SE = \sqrt{s_A^2/n_A + s_B^2/n_B}$, $t = \hat{\Delta}/SE$, with Satterthwaite df and t-quantile interval. Proportion z uses pooled p under the null; its interval uses the separate estimated proportions. Chi-square sums $(O-E)^2/E$, with $E =$ row total times column total divided by N, and df=$(r-1)(c-1)$. Full intermediate values, formulas and alternatives are in methodology_and_results.md and the source functions.

Sensitivity. int_rate: capping at 26.3 gives difference 3.037, CI [3.0172, 3.0568]. dti: capping at 38.47 gives difference 2.2908, CI [2.2545, 2.327].

The cap changes the estimand; it is not a correction of proven errors. Issue-year/term rates are provided to inspect composition, not to claim censoring is resolved.

6. Limitations

Accepted loans exclude rejected applications. Terminal selection excludes unfinished loans and can overrepresent early charge-offs, particularly for newer long-term loans. These are conditional associations, not lifetime default rates or causal effects. Grade and rate reflect related underwriting information; shared borrowers and macroeconomic shocks can violate independence. No reliable borrower identifier was established. Complete cases can be selective, and extreme DTI values require historical documentation.

Holm addresses multiplicity conditional on valid p-values; it does not remove confounding. Subjective practical thresholds should be challenged. Historical file provenance, extraction date and licence are incomplete. The supplied lecture PDFs, quizzes and practice problems were absent, so exact lecture mapping and the 80% syllabus target cannot be verified. Bootstrap and Holm are labelled extensions. No classifier is claimed.

References and code. Supplied CSE558 guidelines/template; supplied preliminary LendingClub profile; the local CSV identified by its saved hash. Historical grade context: LendingClub SEC prospectus (2011), `https://www.sec.gov/Archives/edgar/data/1409970/000095012311058394/c18687e424b3.htm`. This source does not verify definitions for every vintage. See references.md, README.md and src/statistical_tests.py. Replace the code URL and team metadata fields before submission.